

SOFTWARE REQUIREMENTS SPECIFICATION

FOR

RGB-Pose Hub

**A Secure Web Platform for RGB Image Analysis and Hybrid
6DoF Object Pose Estimation**

Prepared by

Abhishu Shaurya (241IT003)

Hradayastha Thakran (241IT034)

Lavish Temani (241IT040)

**DEPARTMENT OF INFORMATION TECHNOLOGY
NATIONAL INSTITUTE OF TECHNOLOGY
KARNATAKA SURATHKAL**

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1. Introduction

1.1 Purpose

RGB-Pose Hub is a secure web-based platform for estimating the full 6DoF pose—rotation and position—of known objects in cluttered industrial scenes, using RGB-D images from datasets like CLUBS. Rather than relying on a single method, it runs two independent pipelines in parallel: a direct regression-based track and a correspondence-based geometric track, then fuses their outputs based on per-object confidence. This hybrid design addresses a key weakness in each approach alone, regression struggles under heavy occlusion, while geometric matching falters with limited visible surface, so combining both improves robustness in real-world clutter. Beyond the core estimation engine, the platform is built for practical, everyday use, with user accounts, a dedicated admin role, and two-factor authentication to protect experiment data, model checkpoints, and credentials. It also includes a standalone RGB image analyser, letting users check image quality, resolution, and basic visual properties independently of running a full experiment. This analyser sits alongside the pose estimation engine as a core feature, not an afterthought, reflecting the platform's dual focus in its name. Altogether, RGB-Pose Hub gives researchers, students, and robotics engineers a single, secure, browser-based environment to upload data, run hybrid pose estimation experiments, analyse images, and track results without touching the underlying code.

1.2 Scope

This project is scoped as a complete, end-user-facing web application built around a hybrid 6DoF pose estimation engine and an RGB image analysis module. It is not meant to be a one-off training script, it's meant to be something multiple users can log into, use, and rely on.

The current implementation covers:

Dataset handling, loading RGB images, depth maps, camera calibration files, and CAD models from the CLUBS dataset, with validation to catch structural issues early.

Preprocessing, resizing, normalization, cropping, and augmentation of RGB-D data before it's fed into either pipeline.

Object detection and segmentation locating and isolating known objects within cluttered scenes.

RGB Image Analyser is a dedicated module that lets users inspect uploaded RGB images independently, checking things like format validity, resolution, brightness/contrast, and general image quality before committing to a full experiment run.

Hybrid pose estimation, made up of two tracks that run in parallel:

Track A: direct 6DoF pose regression using an EfficientPose-based network.

Track B: dense correspondence estimation followed by Perspective-n-Point (PnP) pose recovery, with optional ICP refinement when depth data is available.

Hybrid pose fusion is used for combining Track A and Track B outputs using confidence-based weighting, so the final pose is more robust than either method alone, especially under occlusion.

Evaluation is done by computing standard benchmark metrics (ADD, ADD-S, mAP, 5cm/5° accuracy) so results can be compared meaningfully across experiments.

Visualization and reporting is carried out by overlaying predicted poses on RGB images and generating experiment summaries.

A full web platform layer, user login, a separate admin login with elevated privileges, two-factor authentication, session management, and an experiment dashboard.

Features we're consciously leaving for later versions include multi-camera support, robotic grasp planning integration, cloud deployment at scale, and real-time inference for live warehouse automation. This version is focused on getting the hybrid pipeline, the RGB analyser, and the secure web platform right first

1.3 Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
6DoF	Six Degrees of Freedom representing three rotational and three translational parameters.
ADD	Average Distance metric for pose estimation evaluation.
ADD-S	Symmetric Average Distance metric.
API	Application Programming Interface
CLUBS	Cluttered Boxes Dataset used for evaluating object pose estimation.
CUDA	GPU computing platform developed by NVIDIA.
ICP	Iterative Closest Point algorithm for pose refinement.
mAP	Mean Average Precision.
PnP	Perspective-n-Point algorithm used for pose recovery.
RGB-D	Colour image with depth information.
TensorFlow	Deep learning framework used for model implementation.

1.4 References

1. Oberweger, M. et al. *CLUBS: A Benchmark Dataset for Cluttered Box-Scene 6D Object Pose Estimation*.
Dataset website: <https://clubs.github.io/>
Publication record: https://www.researchgate.net/publication/335987402_CLUBS_An_RGB-D_dataset_with_cluttered_box_scenes_containing_household_objects

2. Bukschat, Y. and Vetter, M. *EfficientPose: Efficient End-to-End 6D Multi-Object Pose Estimation*. <https://arxiv.org/abs/2011.04307>
3. Hodaň, T. et al. *BOP: Benchmark for 6D Object Pose Estimation*. Paper (ECCV 2018): <https://arxiv.org/abs/1808.08319>
4. TensorFlow Documentation. https://www.tensorflow.org/api_docs
5. OpenCV Documentation. <https://docs.opencv.org/>
6. Intel RealSense SDK Documentation. <https://dev.intelrealsense.com/docs>

1.5 Overview

The rest of this document walks through what the RGB-Pose Hub platform actually needs to do and how it's expected to behave. Section 2 gives the bigger picture — what kind of product this is, who's going to use it, and what constraints and assumptions shape its design. Section 3 gets specific: it lays out the functional requirements (including the hybrid pipeline, the RGB image analyser, and the web platform features like login, admin control, and 2FA), the external interfaces, and the non-functional expectations around performance, security, and reliability.

The appendices provide supporting material a workflow diagram showing how data flows through the two pipelines into the fusion step, a glossary of terms, and a version history tracking how this specification evolved as the project's scope was refined.

2. Overall Description

2.1 Product Perspective

RGB-Pose Hub is a secure, web-based research platform designed to estimate the six degrees of freedom (6DoF) pose of known objects in densely cluttered warehouse scenes, while also giving users a way to independently inspect and validate the RGB images they work with. It is not a standalone script, it is a full application with user accounts, an admin role, and login security built in from the start.

The software uses RGB-D images, camera calibration parameters, and CAD models to estimate object position and orientation. Two independent pipelines process each scene in parallel, one using direct pose regression, the other using dense correspondence and PnP and their outputs are fused into a single, more reliable final pose. Alongside this, the RGB image analyser lets users check image quality and properties before committing to a full experiment. The system can serve as a perception module for robotic manipulation, warehouse automation, and industrial inspection use cases.

It is implemented in Python and TensorFlow, using NVIDIA CUDA-enabled GPUs for training and inference, with the CLUBS dataset as the primary benchmark.

2.2 Product Functions

Dataset Handling : Imports RGB images, depth maps, annotations, calibration files, and CAD models, and validates dataset structure before use.

Data Preprocessing : Resizes and normalizes RGB-D images, performs cropping, and applies augmentation (rotation, scaling, colour jittering, occlusion simulation) ahead of training.

RGB Image Analyser : A standalone module where users can upload an RGB image and get back its resolution, format validity, and basic quality indicators (brightness, contrast, blur/noise checks), independent of running a full pose estimation experiment.

Object Detection and Segmentation : Detects visible objects and generates bounding boxes and instance segmentation masks.

6DoF Pose Estimation : Runs two tracks in parallel, Track A does direct pose regression using EfficientPose; Track B does dense correspondence estimation followed by PnP, with optional ICP refinement when depth is available.

Hybrid Pose Fusion : Combines Track A and Track B outputs using confidence-based weighting, so the final pose stays reliable even under heavy occlusion.

Performance Evaluation : Computes ADD, ADD-S, mAP, and 5cm/5° accuracy, and generates experiment reports.

Visualization and Reporting : Overlays predicted poses and CAD models on RGB images, compares against ground truth, and produces experiment summaries and logs.

Dashboard and User Management : Handles user authentication and a separate administrator dashboard, both secured with two-factor authentication (2FA), along with

experiment monitoring, training progress visualization, and activity logging.

2.3 User Classes

The platform is built around four types of users, each with a different reason to log in:

Researchers / ML Engineers use it for model development, training, and benchmarking on CLUBS.

Robotics Engineers consume the estimated poses for manipulation and warehouse automation systems.

Students / Learners use it to learn RGB-D perception and pose estimation hands-on.

Administrators manage user accounts, authentication (including 2FA setup and recovery), experiment logs, and overall system configuration — a role that doesn't exist in a typical single-user research script but is essential once the system is deployed as a multi-user web platform.

2.4 Constraints

The implementation is subject to the following constraints.

- **C.1 Dataset Licensing**

The software shall comply with the licensing conditions of the CLUBS dataset.

- **C.2 Hardware Constraints**

Training requires an NVIDIA CUDA-enabled GPU with at least 8 GB VRAM.

- **C.3 Software Dependencies**

The implementation depends on TensorFlow, OpenCV, NumPy, Pandas, and Matplotlib.

- **C.4 Closed-Set Recognition**

The current implementation supports only the predefined object classes present in the CLUBS dataset.

- **C.5 Camera Calibration**

Pose estimation accuracy depends on correctly calibrated RGB-D cameras.

- **C.6 Experiment Logging**

All experiments shall store checkpoints, training logs, and evaluation metrics.

- **C.7 Application Criticality**

Incorrect pose predictions may affect robotic grasping accuracy and warehouse automation reliability.

- **C.8 Authentication Security**

All user and admin accounts must be protected by 2FA; login sessions must expire appropriately.

2.5 Assumptions and Dependencies

The software assumes the CLUBS dataset is available locally with accurate camera calibration, that Python 3.8+ and CUDA-enabled GPUs are available for training, and that TensorFlow, OpenCV, NumPy, and Pandas are correctly installed. It also assumes CAD models exist for every supported object class, and that users have a valid email or authenticator method available to complete 2FA during login.

3. Specific Requirements

This section specifies the functional and non-functional requirements of RGB-Pose Hub a secure web platform combining RGB image analysis with hybrid 6D object pose estimation.

3.1 Functional Requirements

F1 Load CLUBS Dataset

Description:

The software shall load RGB images, depth maps, camera calibration files, object annotations, and CAD models from the CLUBS dataset while validating the dataset structure. The input to this function consists of the dataset directory, the camera calibration files, and the CAD models. The output is a successfully indexed dataset along with the RGB-D images loaded into memory.

F2 RGB-D Image Preprocessing

Description:

The software shall preprocess RGB-D images through resizing, normalization, cropping and augmentation before model training.

The input to this function consists of RGB images and depth maps. The output is a set of preprocessed RGB-D tensors ready for model training.

F3 Object Detection and Segmentation

Description:

The software shall detect visible objects inside cluttered warehouse scenes and generate bounding boxes together with instance segmentation masks.

The input to this function is an RGB image. The output consists of bounding boxes and instance segmentation masks for the detected objects.

F4 End-to-End Pose Estimation (Track A)

Description:

The software shall estimate the six-degree-of-freedom pose directly using an EfficientPose-based neural network.

The input to this function consists of a cropped object image and the trained model. The output is a rotation matrix and a translation vector representing the estimated 6DoF pose.

F5 Dense Correspondence Estimation (Track B)

Description:

The software shall estimate dense correspondences between image pixels and CAD models followed by Perspective-n-Point pose estimation.

Optional ICP refinement shall be performed whenever depth information is available.

The input to this function consists of an RGB image, a depth map, and a CAD model. The

output is a refined 6DoF pose, obtained after dense correspondence matching and Perspective-n-Point estimation, with optional ICP refinement applied when depth data is available.

F6 Hybrid Pose Fusion

Description:

The software shall combine predictions from Track A and Track B using confidence-based fusion.

The input to this function consists of the Track A prediction and the Track B prediction. The output is the final fused 6DoF pose.

F7 Model Training

Description:

The software shall iteratively train pose estimation models using mini-batches over multiple epochs while storing checkpoints and training statistics.

The input to this function consists of the training dataset and the selected hyperparameters. The output is a trained model, along with the saved checkpoints and training logs.

F8 Model Evaluation

Description:

The software shall evaluate trained models using standard pose estimation metrics.

The input to this function consists of the trained model and the test dataset. The output is an evaluation report, the computed performance metrics (ADD, ADD-S, mAP, and 5cm/5° accuracy), and the corresponding experiment logs.

F9 User Login

Description:

The software shall allow registered users (researchers, robotics engineers, students) to securely log in using a username/email and password, verified with two-factor authentication.

The input to this function consists of the user's username or email, their password, and a 2FA code. The output is an authenticated user session along with access to the dashboard.

F10 Admin Login

Description:

The software shall provide a separate, elevated-privilege login for administrators, also secured with 2FA, granting access to user management, activity logs, and system configuration.

The input to this function consists of the admin credentials and a 2FA code. The output is an admin session, access to the admin dashboard, and the associated user management controls.

F11 Two-Factor Authentication (2FA)

Description:

The software shall generate and verify a time-based one-time code (via authenticator app or email/SMS) as a mandatory second step for both user and admin logins.

The input to this function consists of the registered device or email and the generated one-time password (OTP). The output is a verified login, along with a log of any failed authentication attempts.

F12 Pose Visualization**Description:**

The software shall overlay predicted object poses and CAD models on RGB images for qualitative evaluation.

The input to this function consists of the RGB image, the predicted pose, and the ground truth pose. The output is a pose visualization along with comparison images showing the predicted and ground truth poses side by side.

3.2 External Interface Requirements**3.2.1 User Interfaces**

The software shall be accessible entirely through a web-based dashboard (a CLI may be provided for advanced/batch use).

Core features include dataset upload, RGB image analysis, training configuration, model selection, experiment history, visualization, and secure authentication for both users and admins.

3.2.2 Software Interfaces

Supported Software

- TensorFlow
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Plotly

3.2.3 Communication Interfaces

Communication between frontend and backend shall be performed using REST APIs over HTTP/HTTPS.

The exchanged data shall use JSON while RGB-D images shall be transferred using file uploads.

3.2.4 Hardware Interfaces

Intel/AMD Processor, NVIDIA CUDA GPU, Intel RealSense Camera, USB 3.0, SSD Storage.

3.3 Other Non-Functional Requirements

NF.1 Platform Requirements

- Linux
- Windows
- Python 3.8+

NF.2 Performance

- GPU Acceleration
- Parallel Data Loading
- Low Inference Time
- Scalable Dataset Handling

NF.3 Reliability

The software shall automatically save checkpoints and recover interrupted experiments.

NF.4 Security

- User Authentication
- Password Encryption
- Two-Factor Authentication
- Session Management

NF.5 Maintainability

The software architecture shall remain modular and extensible.

NF.6 Usability

The graphical dashboard shall provide an intuitive interface for researchers and students.

3.4 Design Constraints

- NVIDIA CUDA-enabled GPU recommended.
- TensorFlow framework dependency.
- CLUBS dataset dependency.
- Camera calibration required.
- Closed-set object recognition.

Appendix A

A.1 Block Diagram

The block diagram below illustrates the complete workflow of the proposed Hybrid 6D Object Pose Estimation system. The process begins by loading RGB-D images, depth maps, camera calibration files, and CAD models from the CLUBS dataset. The images are pre-processed before object detection and segmentation. Two complementary pose estimation pipelines estimate the 6DoF pose of each detected object. The predictions are fused to generate the final object pose. Finally, the system evaluates the estimated poses using standard benchmark metrics and visualizes the results through the web dashboard.

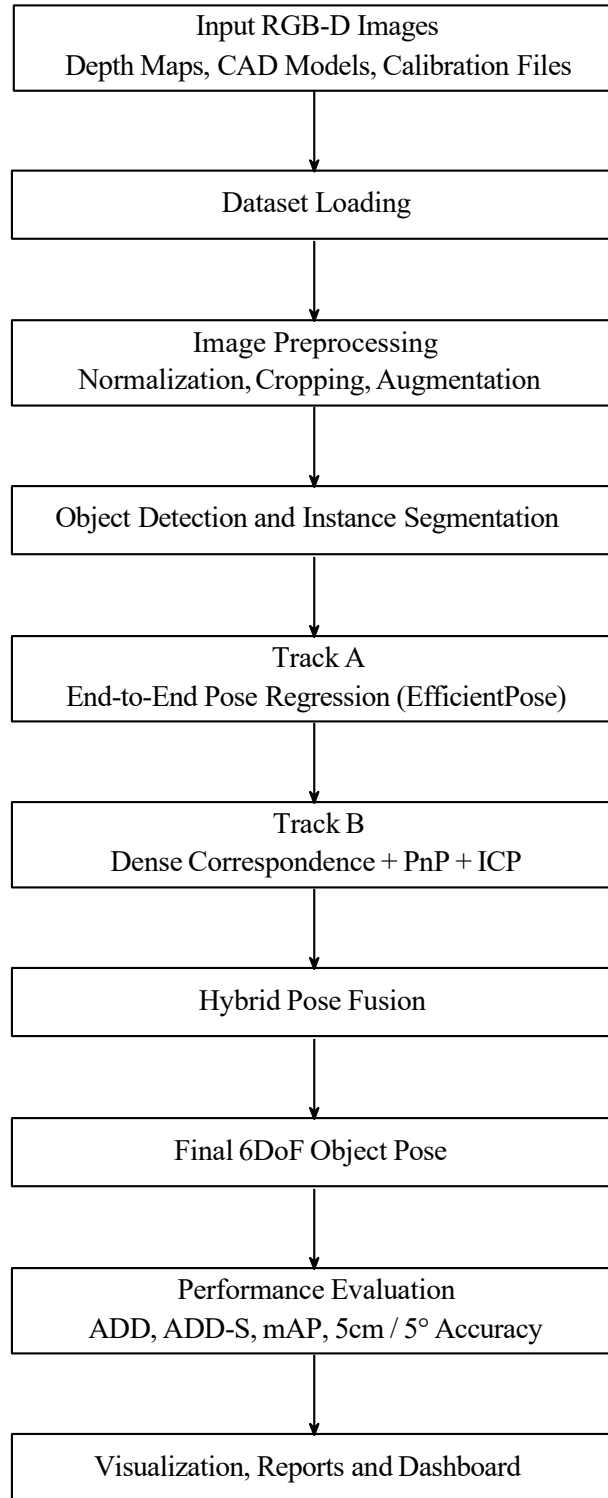


Figure 1: Workflow of the Proposed End-to-End 6D Object Pose Estimation System

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A

ADD — Average Distance Metric

ADD-S — Average Distance for Symmetric Objects

Admin Dashboard — Interface for administrators to manage users, logs, and system configuration

API — Application Programming Interface

Authentication — Process of verifying user identity before granting system access

C

CAD Model — Three-dimensional object representation

CLUBS — Cluttered Boxes Dataset

CUDA — GPU Computing Platform

D

Depth Map — Per-pixel depth information

Dense Correspondence — Pixel to CAD model mapping

E

EfficientPose — Deep learning network used for direct pose regression (Track A)

G

GPU — Graphics Processing Unit

Ground Truth — Reference object pose

H

Hybrid Pose Fusion — Combination of Track A and Track B predictions into a single final pose

I

ICP — Iterative Closest Point

Intel RealSense — RGB-D Camera

M

mAP — Mean Average Precision

P

PnP — Perspective-n-Point

Pose Estimation — Estimation of object rotation and translation

R

RGB Image Analyser, Module for inspecting and validating RGB images independent of the pose estimation pipeline

RGB-D—RGB image with depth

S

Segmentation — Pixel-wise object extraction

Session Management — Handling of active login sessions, including expiry and timeout

Six Degrees of Freedom — Object rotation and translation

T

TensorFlow—Deep Learning Framework

Track A — Direct Pose Regression

Track B — Dense Correspondence Estimation

Two-Factor Authentication(2FA)— Secondary login verification step required for both users and admins

Version

Version	Last Updated	Reason for Change
1.0	02-08-2026	Initial Software Requirements Specification for RGB-Pose Hub.
1.1	02-08-2026	Added functional requirements, external interfaces and dashboard features.
2.0	02-08-2026	Completed Software Requirements Specification with appendix, workflow, index and formatting.

DATA FLOW DIAGRAM

FOR

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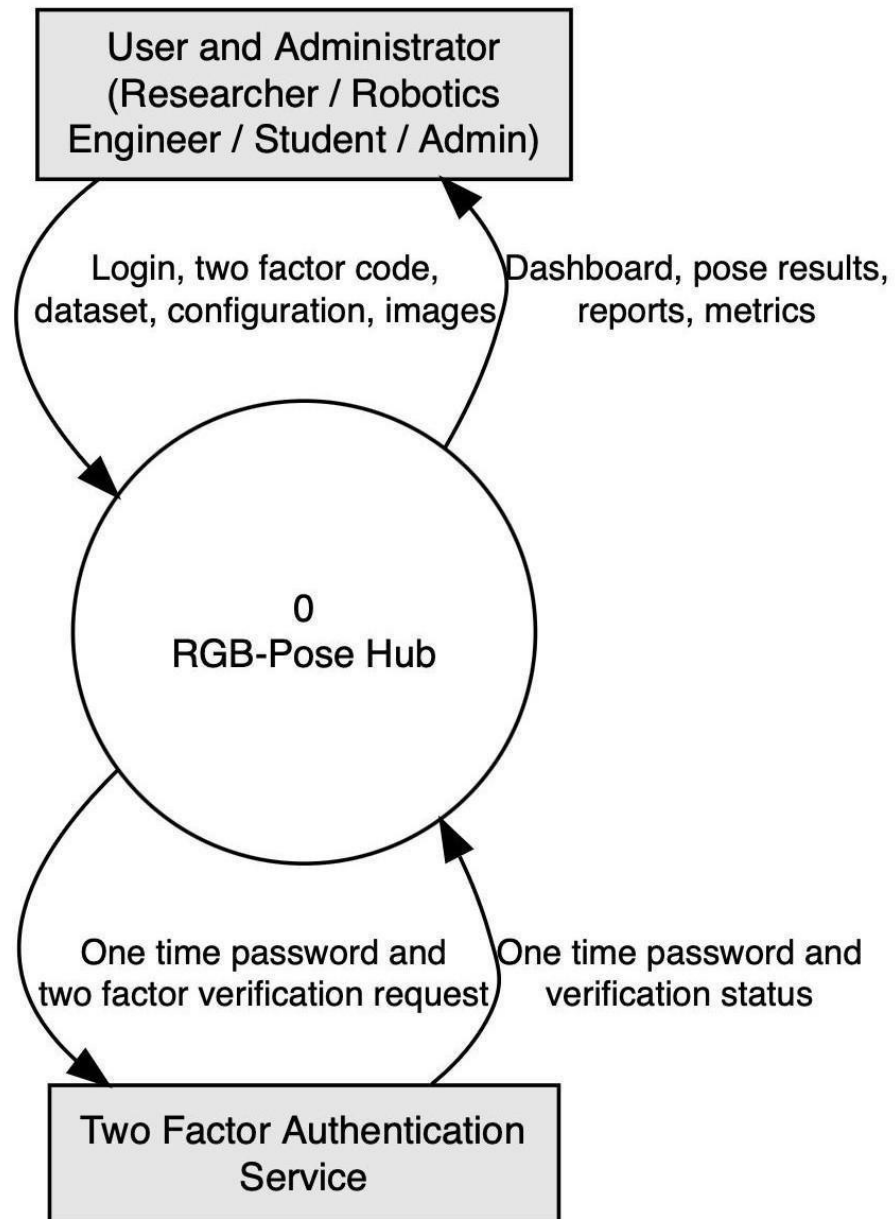
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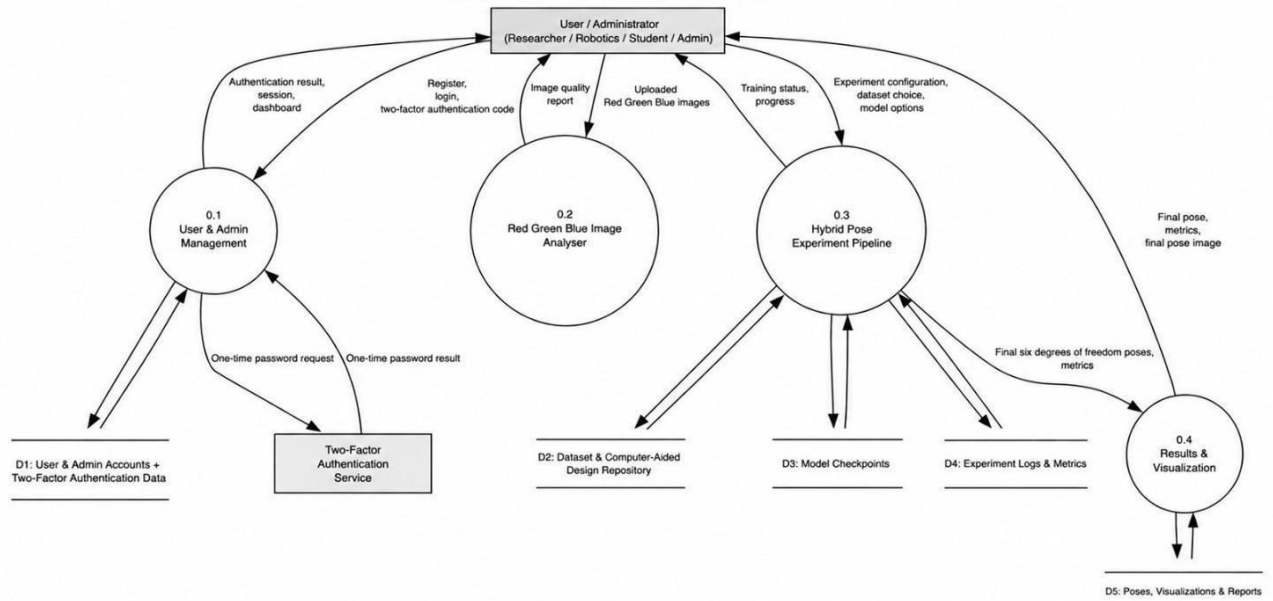
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1. Data Flow Diagram Model

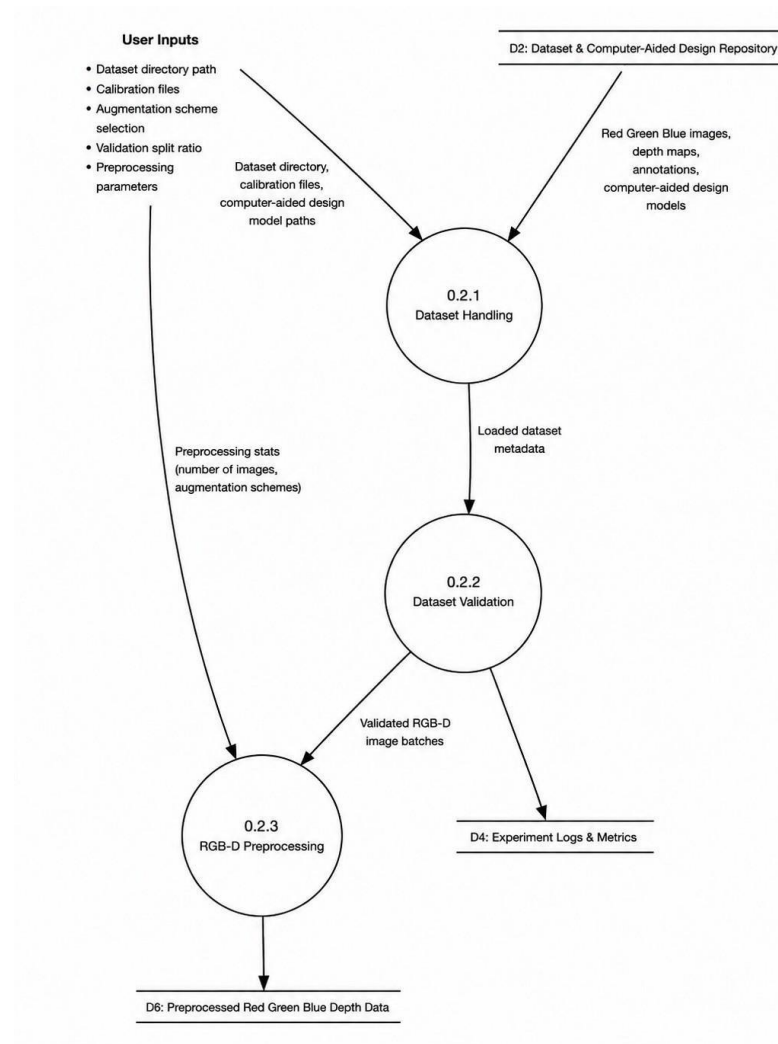
1.1 Level 0 : Context Diagram



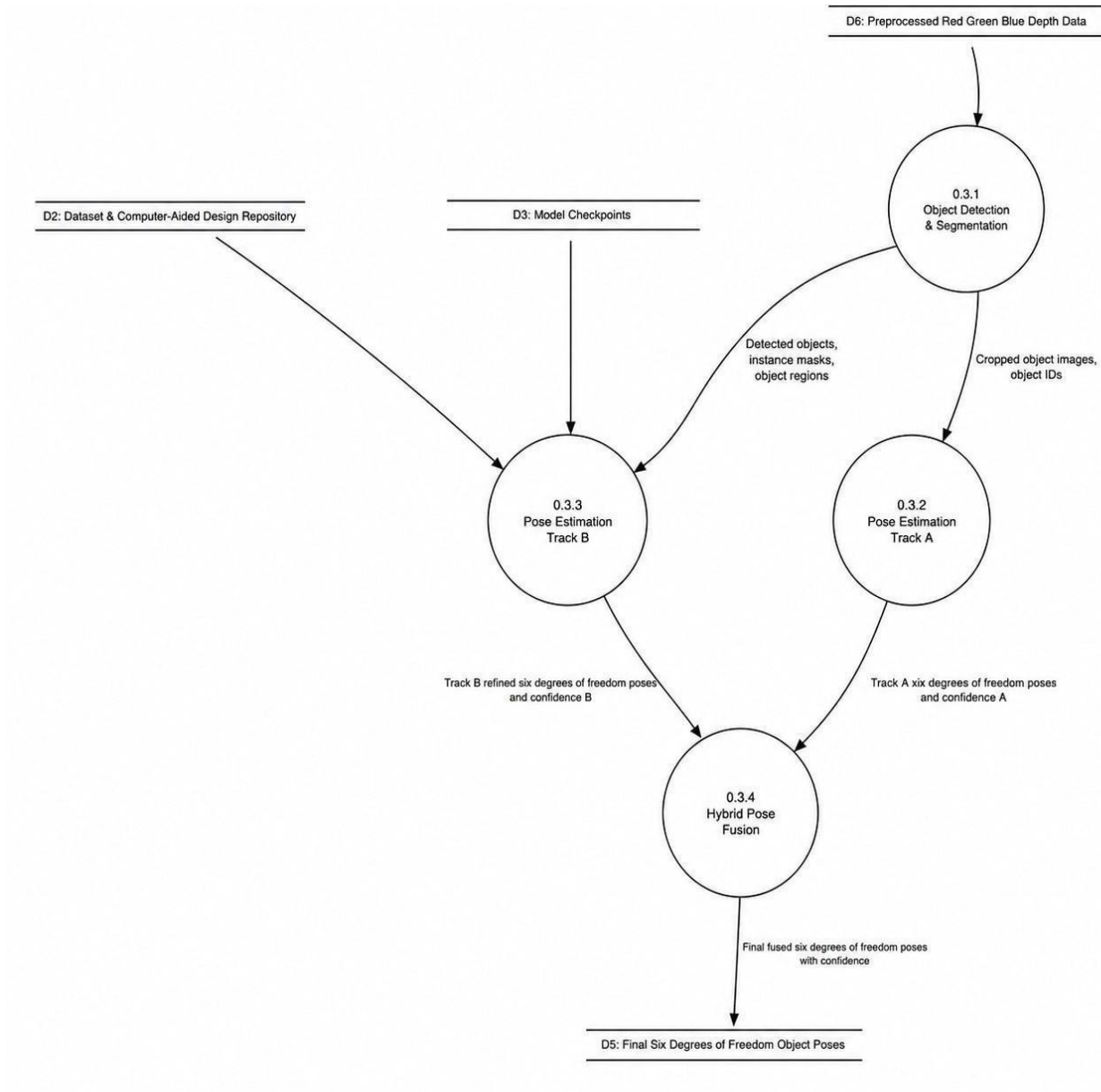
1.2 Level 1 : RGB Pose Hub



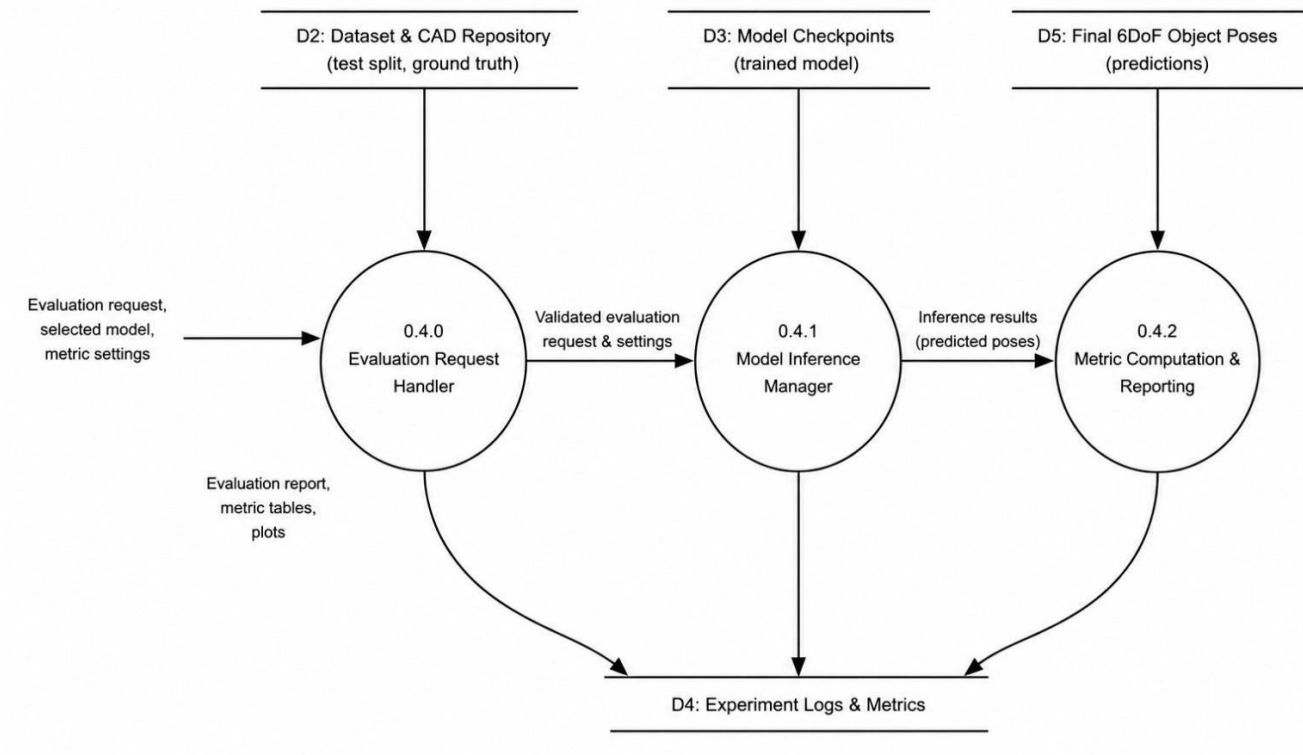
1.3 Level 2A : Dataset Handling and RGB-D Preprocessing



1.4 Level 2B : Detection, Track A , Track B , Hybrid Fusion



1.5 Level 2C : Model Evaluation



Structured Chart

